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1. Introduction and Project Overview

1.1 Executive Summary

The Creator Brand Collaboration System is a platform designed to seamlessly connect digital content Creators with corporate Brands. Identifying the right marketing talent and managing the deal lifecycle is often fragmented. This system provides a centralized hub where Brands can discover Creators through an intelligent recommendation engine, and Creators can manage their portfolios, evaluate proposals, and communicate securely.

1.2 Core System Capabilities

- Profile & Portfolio Management: Users can securely register and establish detailed profiles. Creators upload portfolio media, while a backend Analytics Engine calculates metrics like Engagement Rates and Consistency Scores.

- Intelligent Discovery Feed: A dynamic, swipe-based interface that allows users to explore profiles with ease. The feed is ranked by a Recommendation Engine that calculates match scores based on niche tags and historical metrics.

- Deal Lifecycle Management: A structured workflow that governs how proposals are initiated, sent, accepted, or declined. It utilizes strict state transitions (e.g., from "New" to "Active") trackable via a Deal Dashboard.

- Secure Messaging System: A controlled communication module tied to active deals. Chat threads unlock only upon mutual deal acceptance, keeping negotiations organized and preventing spam.

1.3 Primary Actors

- Creators: Users who register to showcase their portfolios. They interact with the system to manage profiles, view peers in their discovery feed, and respond to incoming deal requests.
- Brands: Business entities seeking marketing talent. Brands utilize the system's recommendation algorithms to discover highly rated Creators and initiate the "Send Deal" workflow.

2. Objectives

2.1 BUSINESS OBJECTIVES

The following is a list of business objectives:

Objective 1: Login and Registration:
Users should be able to make accounts and log in to the platform safely. Users must be able to sign up as either a Creator or a Brand by giving the right information. Authentication makes sure that only people who are allowed to use the system can access its features.

Objective 2: Profile Setup:
Users should be able to make and change their profile information in the system. Brands can set campaign preferences and company details, while creators can upload portfolio content, niche tags, and a personal bio.

Objective 3: Creator's Activities:
The system should let Creators look through niche-based discovery feeds, see other creators who are similar to them, manage their profiles, and respond to requests for collaboration deals.

Objective 4: Brand's Activities:
Brands should be able to find creators through a ranked feed, use filters, make a short list of creators, and start collaboration deals.

Objective 5: Recommendation Engine:
Based on niche compatibility, engagement analytics, and past data, the system will give Brands a list of ranked creator recommendations.

Objective 6: Portfolio Analytics Engine:
The system will look at creator portfolio data and come up with metrics like engagement score, consistency score, and base creator score.

Objective 7: Send Deal:
Brands should be able to make and send collaboration proposals to Creators through the system. These proposals should include information about the campaign, the deliverables, and the deadlines.

Objective 8: Deal Lifecycle:
The system should be able to handle changes in deal status, such as new, accepted, active, declined, expired, completed, and terminated.

Objective 9: Messaging Interaction:
Brands and Creators will only be able to send secure messages to each other after the deal has been accepted.

Objective 10: Dashboard and History:
The system should have dashboards that show active deals, pending proposals, finished collaborations, and past deal records.

2.2 SYSTEM OBJECTIVES

Objective 1: The system shall be developed as a native Android-based mobile application to provide a responsive, mobile-friendly interface for real-time interaction between Creators and Brands.

Objective 2: The system will implement secure user authentication and authorization protocols to ensure that only registered and verified users can access the platform's core features.

Objective 3: The system will utilize a **SQLite database** for the persistent storage of user profiles, portfolio metadata, legal verification documents, deal records, and messaging histories.

Objective 4: The system will integrate an **AI-enhanced recommendation engine** to rank Creators based on niche tags, engagement metrics, and Brand-specific campaign preferences.

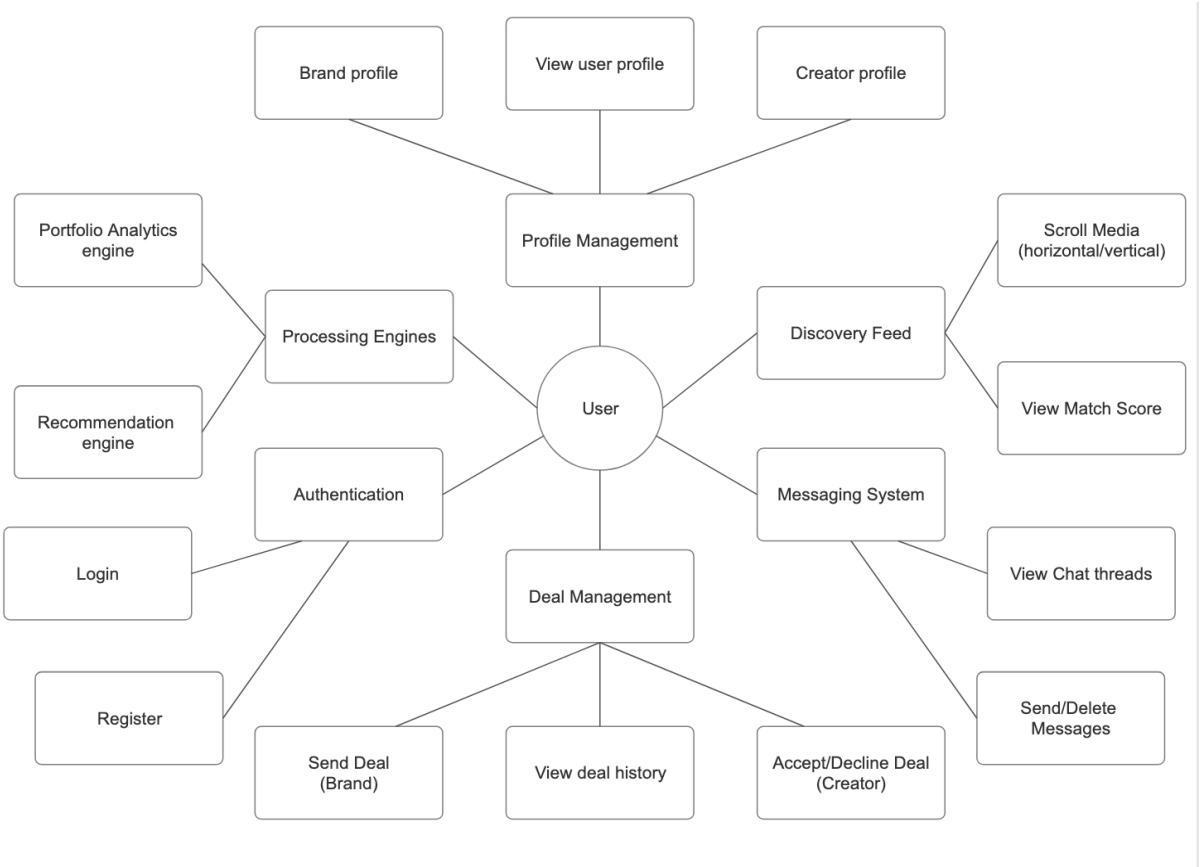
Objective 5: The system will include a **Portfolio Analytics Engine** that utilizes the **YouTube Data API** to calculate objective performance metrics, such as Engagement and Consistency scores.

Objective 6: The system will manage the complete **Deal Lifecycle**, utilizing a strict state-transition model to track collaborations from the "New Proposal" stage through to "Completion" or "Termination."

Objective 7: The system will facilitate a **secure messaging module** that permits the exchange of text and file attachments between Brands and Creators once a deal state is validated as "Active."

Objective 8: The system will provide dynamic **User Dashboards** that query the local database to display the real-time status of active deals, pending proposals, and historical collaboration data.

3. Project Context Diagram



4. Systems Requirements

4.1 "Login and Registration" Requirements

Requirement Title:	Registration and Login
Sequence No:	001
Short description:	Register a new user
Detailed Description:	The landing page presents a single-entry point for all users. The user is prompted to enter their email address and press Continue. The system immediately queries the database to determine whether the email is associated with an existing account: - If the email is not found → proceed to New User Registration - If the email is found → proceed to Returning User Login The user can press: - Continue - Exit screen
Pre-Conditions:	- None
Post Conditions:	- System determines routing path (new user or returning user)
Other attributes:	- Email field must be validated for proper format (e.g., user@domain.com) before submission

Requirement Title:	Registration and Login
Sequence No:	002
Short description:	New User Registration
Detailed Description:	Upon confirming the email is not registered, the system guides the new user through a multi-step registration process: Step 1: Create Password: - User creates a password {must include at least one uppercase letter, at least one number, minimum 8 characters, special characters allowed} Step 2: Personal Information: - Full Name {First and Last} - Phone Number - Security Question and Answer {e.g., What city were you born in? What is your mother's first name?} Step 3: Role Selection: - User selects their role: Creator or Brand Upon completion, the account is created and marked as 'Incomplete.' The system immediately redirects the user to Profile Setup. The user can press: - Next (proceed to next step) - Back (return to previous step) - Cancel (discard all changes) - Exit screen (or close)
Pre-Conditions:	- Email must not already exist in the system
Post Conditions:	- A new user account is created and marked as 'Incomplete' - User is redirected to Profile Setup
Other attributes:	- Duplicate registration is not allowed - Role selection (Creator or Brand) is permanent and cannot be changed after registration

Requirement Title:	Registration and Login
Sequence No:	003
Short description:	Returning User Login
Detailed Description:	Upon confirming the email belongs to an existing account, the system prompts the user to enter their password. Upon successful authentication, the system reads the user's role and routes accordingly: - Brand → redirected to the Video Discovery Feed - Creator → redirected to their Personal Dashboard The user can press: - Submit - Forgot Password (navigates to Seq. 004) - Cancel (discard changes) - Exit screen (or close)
Pre-Conditions:	The user must have already registered - The email must exist in the system
Post Conditions:	- A log of the date and time of login is recorded - User is routed to the correct landing screen based on their role
Other attributes:	- Allow a maximum of 3 login attempts; after the third failed attempt, the system recommends using the 'Forgot Password' feature

Requirement Title:	Registration and Login
Sequence No:	004
Short description:	Forgot Password
Detailed Description:	The user is prompted to provide the following to verify their identity: - Email address - Answer to the Security Question set during Registration (Seq. 002) If both the email address and security answer are correct, the system resets the password and emails a temporary password to the registered email address. The user can press: - Submit - Cancel (discard changes) - Exit screen (or close)
Pre-Conditions:	- The user must have already registered - Email and security question answer must be validated
Post Conditions:	- A temporary password is generated and emailed to the user - System strongly recommends the user change their temporary password upon next login
Other attributes:	- Lock the feature after 3 failed attempts - Consider adding more than one security question for additional verification

4.2 “Profile Setup” Requirements

Requirement Title:	Profile Setup
Sequence No:	001
Short description:	Profile Setup Trigger and Entry Point
Description:	Profile Setup is automatically launched in the following scenarios: - Immediately after a new user completes Registration (Seq. 002), the account is marked 'Incomplete' and the user is redirected here before accessing any other part of the application - When a returning user logs in (Seq. 003) and their account status is still marked 'Incomplete', they are intercepted and redirected to Profile Setup before reaching their role-based landing screen Returning users with a fully 'Complete' account status bypass Profile Setup entirely and are routed directly to their role-based landing screen. The user can press: - Next (proceed through setup steps) - Back (return to previous step) - Save & Exit (save progress, return later; mandatory fields still block app entry) - Cancel (discard unsaved changes on current step)
Pre-Conditions:	- User must have a registered account marked as 'Incomplete'
Post Conditions:	- Once all mandatory fields are submitted, account status is updated from 'Incomplete' to 'Complete' - User is routed to their role-based landing screen (Brand → Discovery Feed; Creator → Dashboard)
Other attributes:	- Users cannot navigate to any other part of the application while their profile is 'Incomplete,' except to log out Progress is saved between sessions; users who exit mid-setup will resume from where they left off

Requirement Title:	Profile Setup
Sequence No:	002
Short description:	Common Profile Information; Both Creators and Brands
Description:	Regardless of role, all users complete the following shared profile fields in the first step of Profile Setup: - Profile Photo; optional. - Motto / Headline; mandatory. - Bio; optional; a longer description of the user or brand. - Social Media Link (YouTube); After completing this step, the user is branched into their role-specific setup: - Creator → proceeds to Seq. 007 (Creator-Specific Setup) - Brand → proceeds to Seq. 008 (Brand-Specific Setup) The user can press: - Next (proceed to role-specific setup) - Back (return to previous step) - Save & Exit (save progress and return later) - Cancel (discard unsaved changes on current step)
Pre-Conditions:	- User must have passed the entry trigger in Seq. 005
Post Conditions:	- Saved common profile fields are stored and reflected on the user's public profile immediately - User is branched to their role-specific setup screen
Other attributes:	- Motto is the only mandatory field in this step; the user cannot advance to role-specific setup without it - Profile photo defaults to a generic avatar placeholder if none is uploaded

Requirement Title:	Profile Setup
Sequence No:	003
Short description:	Creator-Specific Profile Setup
Description:	After completing the common profile fields (Seq. 006), Creators complete the following additional steps: Step 1 - Niche Tags - Creator selects at least 1 and up to 5 Niche Tags that represent their content focus (e.g., Fitness, Tech, Beauty, Gaming, Travel, Fashion, Food) - Creator enters their Location - City, State/Region, and Country (e.g., Dallas, Texas, USA); mandatory - Creator selects their Primary Audience Region(e.g., US - Southwest, UK, Southeast Asia) Step 2 - Video Uploads: - Creator uploads at least 1 and up to 10 videos to their portfolio The user can press: - Submit (complete Profile Setup and enter the main application) - Back (return to common info step) - Save & Exit (save progress and return later; mandatory fields still block app entry) - Cancel (discard unsaved changes on current step)
Pre-Conditions:	- User must have completed Seq. 006 (Common Info) - User's registered role must be 'Creator'
Post Conditions:	- Creator's niche tags and video portfolio are saved and visible on their public profile - Account status is updated to 'Complete' once all mandatory fields are satisfied - Creator is redirected to their Personal Dashboard
Other attributes:	- Niche Tags influence the Creator's ranking in the algorithmic match engine; selecting accurate tags is strongly recommended

Requirement Title:	Profile Setup
Sequence No:	004
Short description:	Brand-Specific Profile Setup
Description:	After completing the common profile fields (Seq. 006), Brands complete the following additional steps: Step 1 - Industry Tags: - Brand selects at least 1 Industry Tag that represents their business sector (e.g., Sportswear, Consumer Tech, Food & Beverage, Healthcare, Finance, Beauty & Cosmetics) - Target location tag Step 2 - Campaign Description: - Brand creates one or more Campaigns, each consisting of a Campaign Name (mandatory per entry), and a Campaign Brief / Description (mandatory per entry) - Brand can add, edit, or delete Campaigns at any time from their profile settings The user can press: - Submit (complete Profile Setup and enter the main application) - Back (return to common info step) - Save & Exit (save progress and return later) - Cancel (discard unsaved changes on current step)
Pre-Conditions:	- User must have completed Seq. 006 (Common Info) - User's registered role must be 'Brand'
Post Conditions:	- Brand's Industry Tags and Campaign Description are saved and reflected on their public profile - Account status is updated to 'Complete' once at least 1 Industry Tag is saved - Brand is redirected to the Discovery Feed
Other attributes:	- Industry Tags are the only mandatory field in this step; Campaign Description is optional and can be added or edited later from the Brand's profile settings - Brands may select multiple Industry Tags if their campaigns span more than one sector

4.3 "CREATOR'S ACTIVITIES" REQUIREMENTS

Requirement Title:	Creator's Activities
Sequence No:	001
Short description:	Manage Creator Profile
Detailed Description:	After logging in, creators must be able to see and control their profile. The system lets creators see their profile information and make changes to things like their bio, personal information, and other profile-related settings. This is in line with the UML flow, which has creators go from "Manage Profile" to "View Profile" and then to "Edit Profile." The user can press: - View Profile - Edit Profile - Save Changes - Exit Screen
Pre-Conditions:	- Creator must be logged into the system
Post Conditions:	- Updated profile information is saved and reflected in system.
Other attributes:	- None

Requirement Title:	Creator’s Activities
Sequence No:	002
Short description:	Explore Creator Feed
Detailed Description:	Creators need to be able to look through a niche-based discovery feed that shows other creators and their work. The system lets creators scroll through the feed to look at profiles and find other creators who work in the same area as them. The system lets you scroll vertically all the time so you can easily look at a lot of creators at once. Creators can also choose a creator from the feed to look at their profile and content for ideas and to see what's popular. This fits with the UML steps "Scroll Niche Feed" and "View Similar Creators," which are combined into one continuous flow of interactions. The user can press: - Scroll through feed - View Creator Profile - View Creator Content - Save Content - Exit Screen
Pre-Conditions:	- The user must be logged into the system.
Post Conditions:	- Creators can explore and interact with content from similar creators.
Other attributes:	- Content may be ranked based on engagement or relevance.

Requirement Title:	Creator's Activities
Sequence No:	003
Short description:	Review and Respond to incoming Deals
Detailed Description:	Creators need to be able to see deal requests from brands and respond to them. The system shows collaboration proposals in a structured list, so creators can look over the terms of the deal before deciding. Creators can choose to accept or turn down a deal right from the request interface. If the deal is accepted, it goes into an active state and you can send messages. If you say no to the deal, it is no longer active and is stored in deal history. The user can press: - View Deal Details - Accept - Decline
Pre-Conditions:	- Creator must be logged into the system - At least 1 deal request must exist.
Post Conditions:	- Deal status is updated based on the creator's response.
Other attributes:	- None

Requirement Title:	Creator's Activities
Sequence No:	004
Short description:	Access Messages Through Deal History
Detailed Description:	People who make things must be able to see past deals and messages about those deals. The system sorts conversations by deal history, so messages are linked to certain collaborations. The user can press: - View Deal History - Open Message Thread - Send Message - Exit Screen
Pre-Conditions:	- Creator must have deal history records.
Post Conditions:	- Creators can view and participate in conversations tied to deals.
Other attributes:	- Messaging follows active deal constraints.

4.4 “BRAND’S ACTIVITIES” REQUIREMENTS

Requirement Title:	Brand’s Activities
Sequence No:	001
Short description:	Browse Creator Discovery Feed
Detailed Description:	Brands should be able to look through a discovery feed that shows creators who are open to working with them. The system will give creators a score based on how well they match the brand's campaign preferences and the creator's profile data. Brands can scroll through creator cards and see summary information before opening full profiles in the discovery feed. Brands need to be able to scroll vertically all the time in the discovery feed so they can easily look through multiple creators. The system may also let users navigate horizontally within creator cards to see multiple media items, like portfolio content, without leaving the feed. Brands can initiate a deal proposal from any creator profile within the system if they want to collaborate with them. The user can press: - Scroll Vertically - Navigate Horizontally - View Creator - Save Creator - Send Deal - Exit Screen
Pre-Conditions:	- Brand must be logged into the system.
Post Conditions:	- Selected creator profiles are opened for detailed viewing
Other attributes:	- Ranking may consider engagement rate, category alignment, and previous collaboration ratings.

Requirement Title:	Brand's Activities
Sequence No:	002
Short description:	Search creators using filters
Detailed Description:	Brands should be able to look for creators using a variety of filters. Filters help brands narrow down results by things like content category, number of followers, location, engagement rate, and type of platform. When you use filters, the search results change in real time. The user can press: - Apply Filter - Reset Filter - View Profile - Save Creator - Send Deal
Pre-Conditions:	- Brand must be logged into the system.
Post Conditions:	- System displays filtered creator results that match the selected criteria.
Other attributes:	- Filters should support multiple simultaneous conditions.

Requirement Title:	Brand's Activities
Sequence No:	003
Short description:	Save Creators to Shortlist
Detailed Description:	Brands need to be able to add creators to a shortlist for later consideration. The shortlist helps companies get ready for future campaigns by organizing possible partners. You can look over saved creators later and choose them when you send deal proposals. The user can press: - Add to Shortlist - Send Deal - Remove from Shortlist - View Shortlist - View Creator Profile
Pre-Conditions:	- Brand must be logged into the system
Post Conditions:	- Selected creators are saved in the brand's shortlist list.
Other attributes:	- No limit on number of people can be shortlisted

4.5 "RECOMMENDATION ENGINE" REQUIREMENTS

Requirement Title:	Recommendation Engine
Sequence No:	001
Short description:	Generate ranked creator recommendations for brands as a background matching process.
Detailed Description:	The system shall include a Recommendation Engine that operates as an internal background matching process. The Recommendation Engine processes the available creator data, portfolio analytics data, niche tags, and brand reference data in order to calculate match score between brand and creators. The Recommendation Engine must then rank the eligible creators based on match score and results must be presented through the user interface by displaying creator videos according to their ranking scores.
Pre-Conditions:	- A Brand profile must exist in the system. - Creator profile data must exist in the database. - Relevant brand preference data and creator-related data must be available to the system. - Portfolio analytics data should be available if used as part of recommendation scoring.
Post Conditions:	- Match scores are calculated for eligible creators. - A ranked list of recommended creators is generated. - Recommendation results may be stored temporarily for faster retrieval.
Other attributes:	- The system should refresh recommendations when creator data or brand preferences change. - The prototype may use simplified recommendation logic or scoring criteria. - The system should return no recommendation results when sufficient matching data is unavailable.

4.6 "PORTFOLIO ANALYTICS ENGINE" REQUIREMENTS

Requirement Title:	Portfolio Analytics
Sequence No:	001
Short description:	Analyze creator portfolio data in background to generate insights.
Detailed Description:	The system must have Portfolio Analytics Engine that operates as an internal background process. Portfolio Analytics Engine evaluate creator portfolio data, feed items and profile related data like total views, likes, post frequency, and past promotion counts in order to generate insights about creator like: - Consistency Score (0-100) that represents audience interaction strength - Engagement Score (0-100) that represents how regularly the creator posts content over a defined time period. - Overall Base Score (0-100) which be calculated as: Base Creator Score = (0.50 × Engagement Score) + (0.50 × Consistency Score) Generated analytics results may be display on the creator profile or may also be used internally by the Recommendation Engine to support creator-brand matching.
Pre-Conditions:	- A creator profile must exist in the system. - Portfolio items and related data must already be stored in the database. - The Portfolio Analytics Engine must have access to the required creator data.
Post Conditions:	- Analytics results may be passed to the Recommendation Engine.
Other attributes:	- Analytics should be recalculated whenever the creator updates portfolio items or after certain interval of time.

4.7 "SEND DEAL" REQUIREMENTS

Requirement Title:	Send Deal
Sequence No:	001
Short description:	Brand sends collaboration proposal
Detailed Description:	Brands need to be able to send creators offers to work together. The proposal should include information about the campaign, such as what will be delivered, how much will be paid, when it will happen, and any other requirements for working together. When a brand sends a deal, the system lets the brand give a Quick Deal Overview. The overview shows important information about the campaign, like the title, so the creator can get the main points of the proposal without having to open the full deal. This helps creators quickly sort through incoming proposals and choose which ones to look at more closely. The system must automatically let the chosen creator know through in-app alerts, push notifications, or email once the proposal is sent. This makes sure that the creator knows right away about the new chance to work together. The system then saves the proposal and adds it to the dashboards of both the brand and the creator so they can keep track of it and act. The user can press: - Send Deal - Add a Campaign Title - Save Draft - View Deal - Cancel - Exit Screen
Pre-Conditions:	- Brand must select a creator profile.
Post Conditions:	- Deal proposal is stored in the system and delivered to the creator's dashboard.
Other attributes:	- Proposal must include required campaign details before submission.

4.8 “DEAL LIFECYCLE” REQUIREMENT

Requirement Title:	Deal Lifecycle
Sequence No:	001
Short description:	Initialize Deal and Notify Creator
Detailed Description:	When a deal is initialized by the Brand, the system must immediately notify the Creator via push or email notification. In the Creator's Deal Dashboard, display the incoming proposal clearly under a "New Deals" section. Provide the Creator with actionable options to either Accept or Decline the proposal. For the Brand's dashboard, display the proposal under a "Sent Deals" or "Pending" section so they can track its status. The user can press: - Accept (Creator) - Decline (Creator) - Withdraw/Cancel (Brand)
Pre-Conditions:	- Brand submits a formal proposal.
Post Conditions:	- Deal is logged in both respective dashboards (New Deals/Sent Deals). - Notifications are dispatched.
Other attributes:	- None.

Requirement Title:	Deal Lifecycle
Sequence No:	002
Short description:	Process Creator Decision (Accept/Decline/Expire)
Detailed Description:	**If Accepted:** Transition the deal status to "Accepted" and allow messaging. Once the conversation officially begins in the DMs, transition the deal to the "Active" state. **If Declined:** Transition the deal status to "Declined" and move it to the "Deal History" dashboard. Provide an optional dropdown for the Creator to give a reason for decline (e.g., "Schedule Full", "Budget too low") to give the Brand helpful feedback. **If Ignored:** Automatically track the timestamp of the proposal. If no action is taken for more than a week (7 days), automatically move the deal to the history dashboard with an "Expired" status to prevent dashboard clutter. The user can press: - Submit Reason (if declining)
Pre-Conditions:	- Deal must be in the "New Deals" / pending state.
Post Conditions:	- Deal status is officially updated. - Messaging is unlocked (if accepted) or deal is archived (if declined/expired).
Other attributes:	- None.

Requirement Title:	Deal Lifecycle
Sequence No:	003
Short description:	Close and Archive Deal
Detailed Description:	Active deals can be officially closed after the collaboration ends. Require a "mutual close" (e.g., Brand clicks "Complete Deal" and Creator confirms it) to ensure both parties agree the work is done. **Early Termination:** Provide an option to cancel/terminate an active deal if the collaboration falls apart mid-project, moving it to history as "Terminated" rather than "Completed". After a successful closing, prompt both parties to rate each other and write comments. Publish these reviews on their respective profiles under a public "Reviews" section and feed this data into the system to recalculate their overall "Aura Score". The user can press: - Complete Deal - Confirm Close - Terminate Deal - Submit Review
Pre-Conditions:	- Deal must be in the "Active" state.
Post Conditions:	- Deal is archived to history. - Reviews are published and Aura Scores are updated.
Other attributes:	- Requires two-party confirmation for a successful completion status.

4.9 “MESSAGING INTERACTION” REQUIREMENTS

Requirement Title:	Messaging Interaction
Sequence No:	001
Short description:	Secure Chat for Active Deals
Detailed Description:	Users can only message each other if the deal status is currently "Active". - The system will: - Deliver messages in real-time so users don't have to refresh the page to see new replies. - Send a push or email notification to the recipient when a new message arrives. - Show message status as "Sent" (when stored in the database) and "Delivered" (when pushed to the recipient). The user can press: - Send - Cancel (discard message) - Exit Screen (or close)
Pre-Conditions:	- The deal status must be validated as "Active".
Post Conditions:	- Message is delivered in real-time and a notification is sent to the recipient.
Other attributes:	- None

Requirement Title:	Messaging Interaction
Sequence No:	002
Short description:	File Sharing and Media Attachments
Detailed Description:	Allow users to attach files directly in the chat thread. Creators and Brands can send contracts, media kits, or draft videos for approval. The user can press: - Add Attachments - Send - Cancel (discard upload) - Exit Screen (or close)
Pre-Conditions:	- The user must be in an active chat thread.
Post Conditions:	- Save all shared files securely to that specific deal's history for easy access later.
Other attributes:	- - Support standard media formats (PDF, MP4, JPG, PNG).

4.10 "Deal Dashboard and History" Requirements

Requirement Title:	Deal Dashboard and History
Sequence No:	001
Short description:	Active Tab - Current Workspace
Description:	The Active Tab is the primary workspace of the Deal Dashboard. It displays all deals that are currently in progress between a Creator and a Brand. Each deal card displays the following information: - Brand name and Creator name - Current deal status (e.g., In Progress, Awaiting Deliverable, Under Review) Messaging (DM) is unlocked exclusively for deals appearing in the Active Tab. Deals in the Pending or History tabs do not have messaging access. The user can press: - Open Deal - view full deal details and deliverables - Open DM - launch the direct messaging thread for that deal - Filter - apply sort/filter options
Pre-Conditions:	- User must be logged in with a 'Complete' profile - At least one deal must have been accepted and moved to Active status
Post Conditions:	- All interactions within a deal (messages, status updates) are logged with a timestamp
Other attributes:	- Both Creators and Brands see the Active Tab, but the deal cards are displayed from their respective perspectives

Requirement Title:	Deal Dashboard and History
Sequence No:	002
Short description:	Pending Tab - Incoming and Outgoing Offers
Description:	The Pending Tab displays deals that are awaiting a response. The view differs by role: Creator View: - Displays all new incoming deal offers sent by Brands that the Creator has not yet responded to - Each card shows: Brand name, offer summary, date sent, and a countdown timer showing days remaining before the offer expires (7-day limit) - Creator can Accept or Decline each offer directly from this tab Brand View: - Displays all deal offers the Brand has sent that have not yet received a reply from the Creator - Each card shows: Creator name, offer summary, date sent, and the remaining time before the offer expires - Brand can view offer details or retract the offer The user can press: - Accept - move the deal to the Active Tab (Creator only) - Decline - reject the offer; Creator is prompted to optionally provide a reason - View Details - expand the full offer details - Retract Offer - cancel the sent offer before it is responded to (Brand only) - Filter - apply sort/filter options
Pre-Conditions:	- A Brand must have sent a deal offer to a Creator - The offer must not have already expired, been accepted, or been declined
Post Conditions:	- If Accepted: deal is moved to the Active Tab for both the Creator and the Brand - If Declined: deal is moved to the History Tab under 'Declined' status; the Creator's reason (if provided) is stored and visible to the Brand - If Retracted: deal is removed from both users' dashboards entirely
Other attributes:	- Offers that receive no response within 7 days are automatically moved to the History Tab under 'Expired' status by the system.

Requirement Title:	Deal Dashboard and History
Sequence No:	003
Short description:	History Tab - Deal Archive
Description:	The History Tab is a read-only archive of all deals that have reached a terminal state. Deals are organized into three sub-states: Completed: - Deals that ran their full course and were successfully concluded by both parties - Full deal details, deliverables, and message history are preserved and accessible Declined: - Deals that the Creator explicitly rejected via the Pending Tab - The Creator's decline reason (if provided) is displayed on the deal card and visible to the Brand Expired: - Deals that sat in the Pending Tab for 7 consecutive days without any action from the Creator - The system automatically moves these deals to Expired status - no manual action is required from either party The user can press: - View Details - expand full deal record (read-only) - Filter - apply sort/filter options (see Seq. 012)
Pre-Conditions:	- A deal must have reached one of the three terminal states: Completed, Declined, or Expired
Post Conditions:	- All archived deals are permanently retained in the History Tab and cannot be deleted by the user - No further actions (messaging, editing, or status changes) can be taken on deals in the History Tab
Other attributes:	- Both the Creator and the Brand retain their own copy of the deal record in their respective History Tabs - Completed deals may be referenced by the system in future analytics or performance summaries

Requirement Title:	Deal Dashboard and History
Sequence No:	004
Short description:	Filters and Organization
Description:	The system provides filtering and sorting controls available across all three tabs (Active, Pending, and History). Sort Options: - Date - sort by most recent or oldest first - Status - group deals by their current status label (e.g., In Progress, Awaiting Deliverable, Completed, Declined, Expired) Filter Options: - By Role-Specific Party - Creators can filter by Brand name; Brands can filter by Creator name - By Date Range - narrow results to a specific time window - By Status - show only deals matching a selected status The user can press: - Apply - apply the selected sort or filter options - Reset - clear all active filters and return to the default view
Pre-Conditions:	- User must be on the Deal Dashboard with at least one deal present in any tab
Post Conditions:	- The selected filter or sort is applied immediately and persists for the duration of the session - Filters reset to default when the user navigates away from the Dashboard
Other attributes:	- Active filters are visually indicated so the user is always aware that a filter is applied - Filters are independent per tab - applying a filter on the Active Tab does not affect the Pending or History tabs

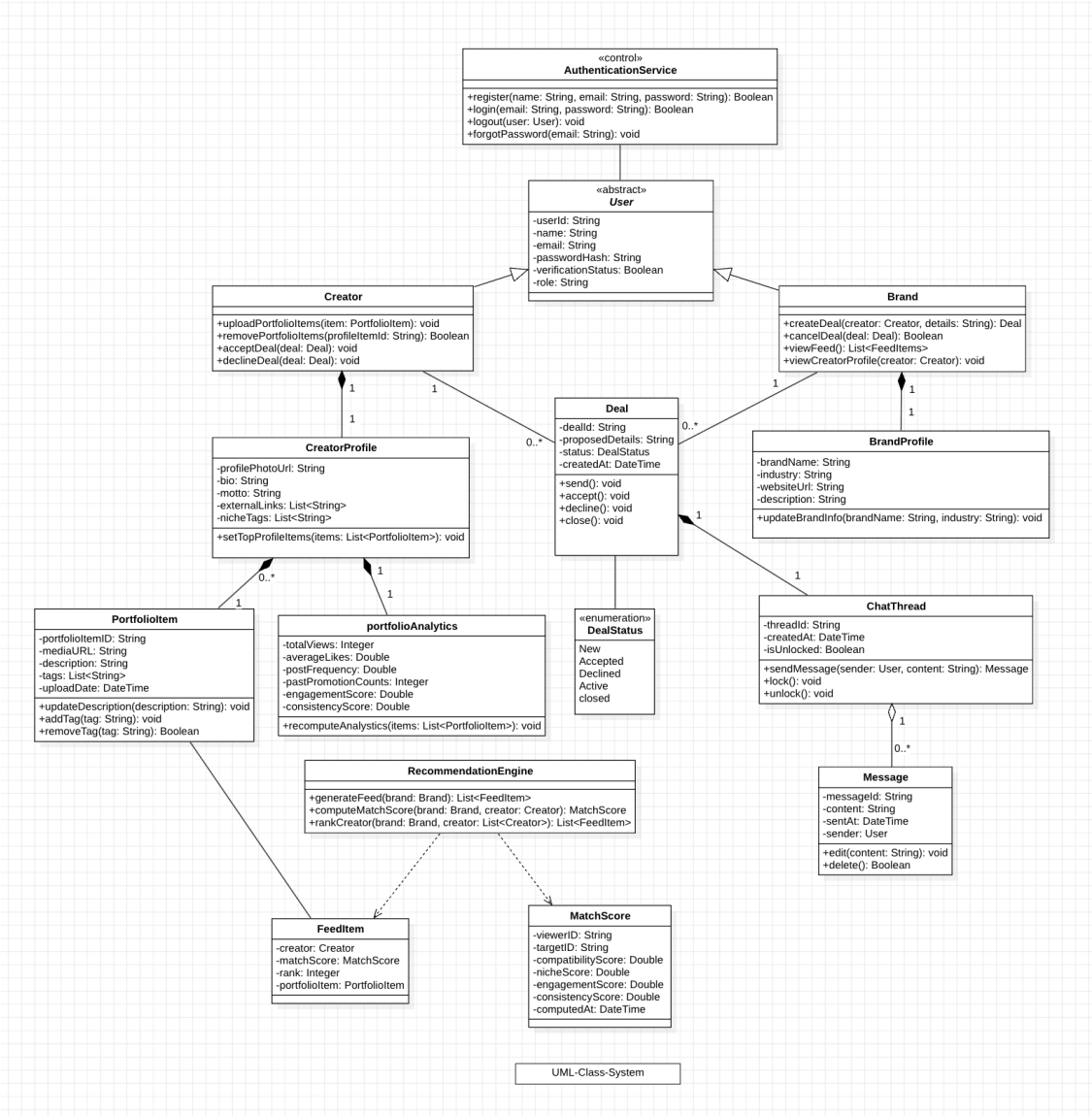
5. Software Processes and Infrastructure

5.1 Hardware and Infrastructure

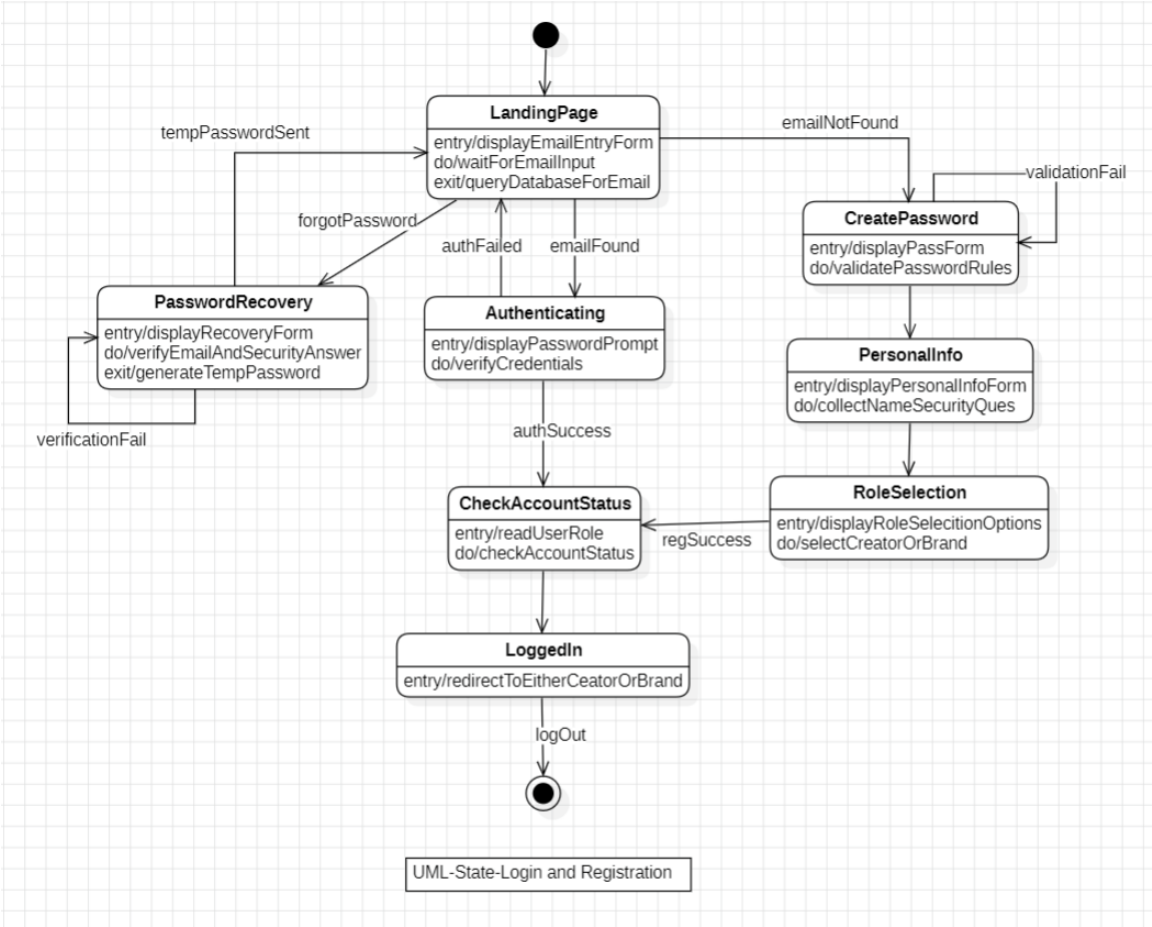
The system will be implemented as an Android-based mobile application running on Android smartphones.

5.2 UML Diagrams

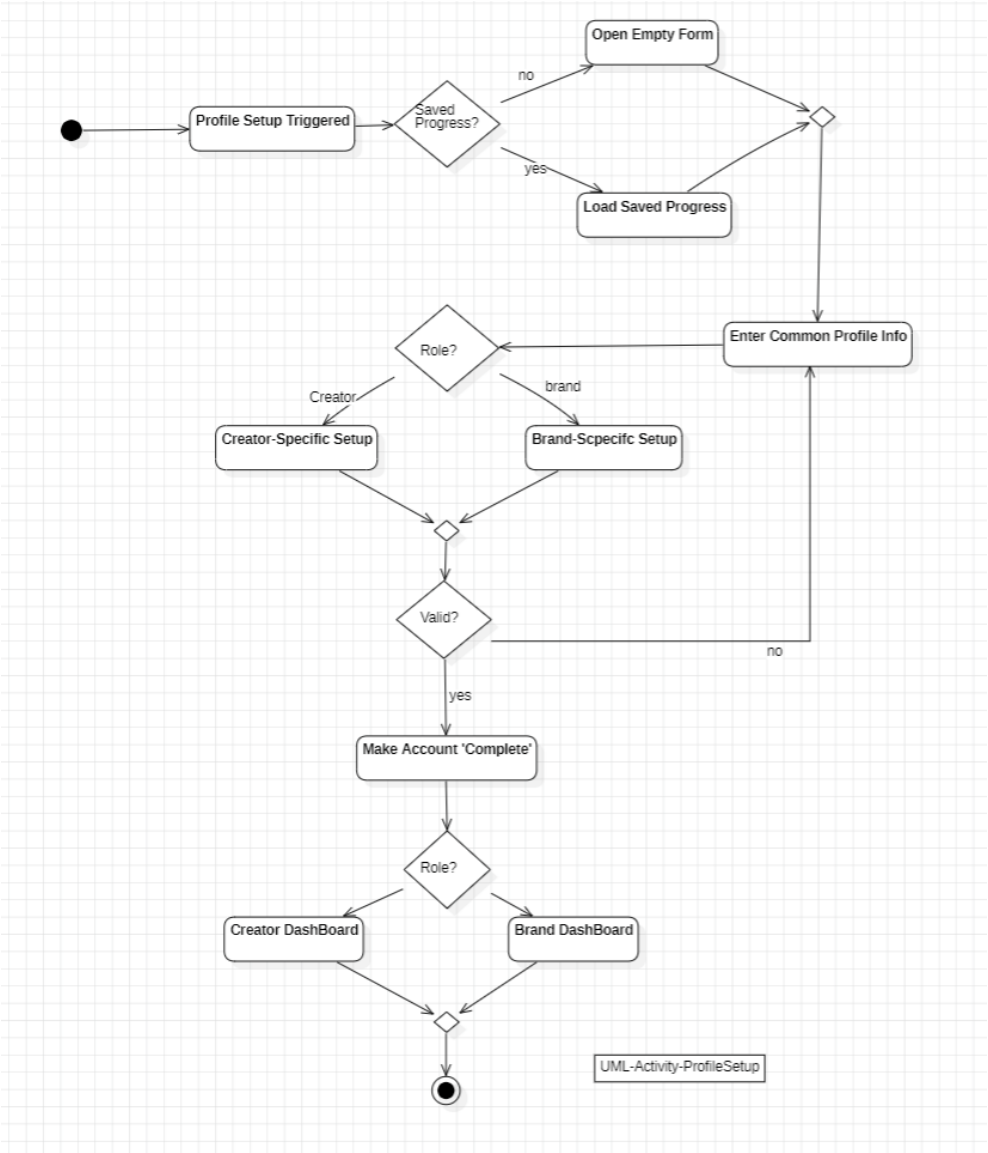
UML Diagram: “Class- System”



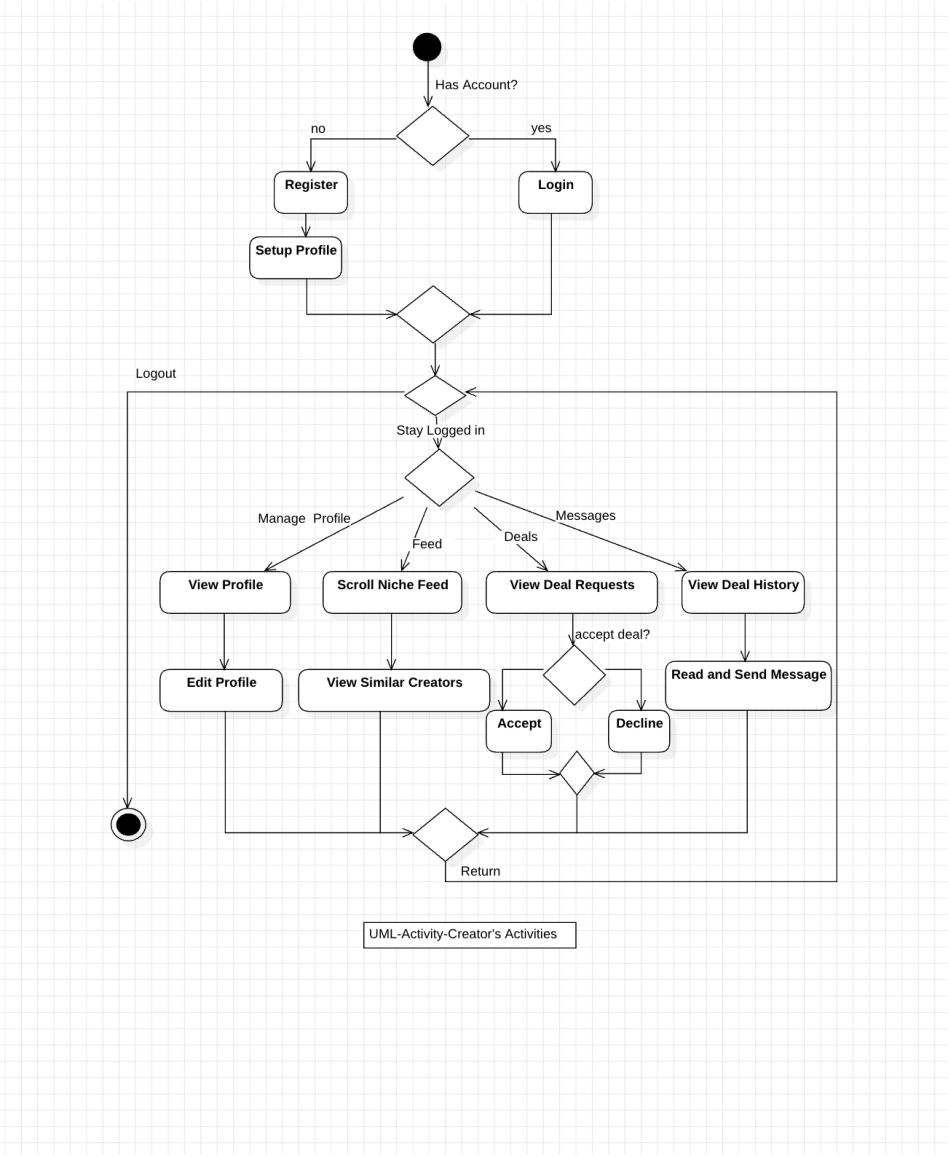
UML DIAGRAM: “STD - LOGIN AND REGISTRATION”



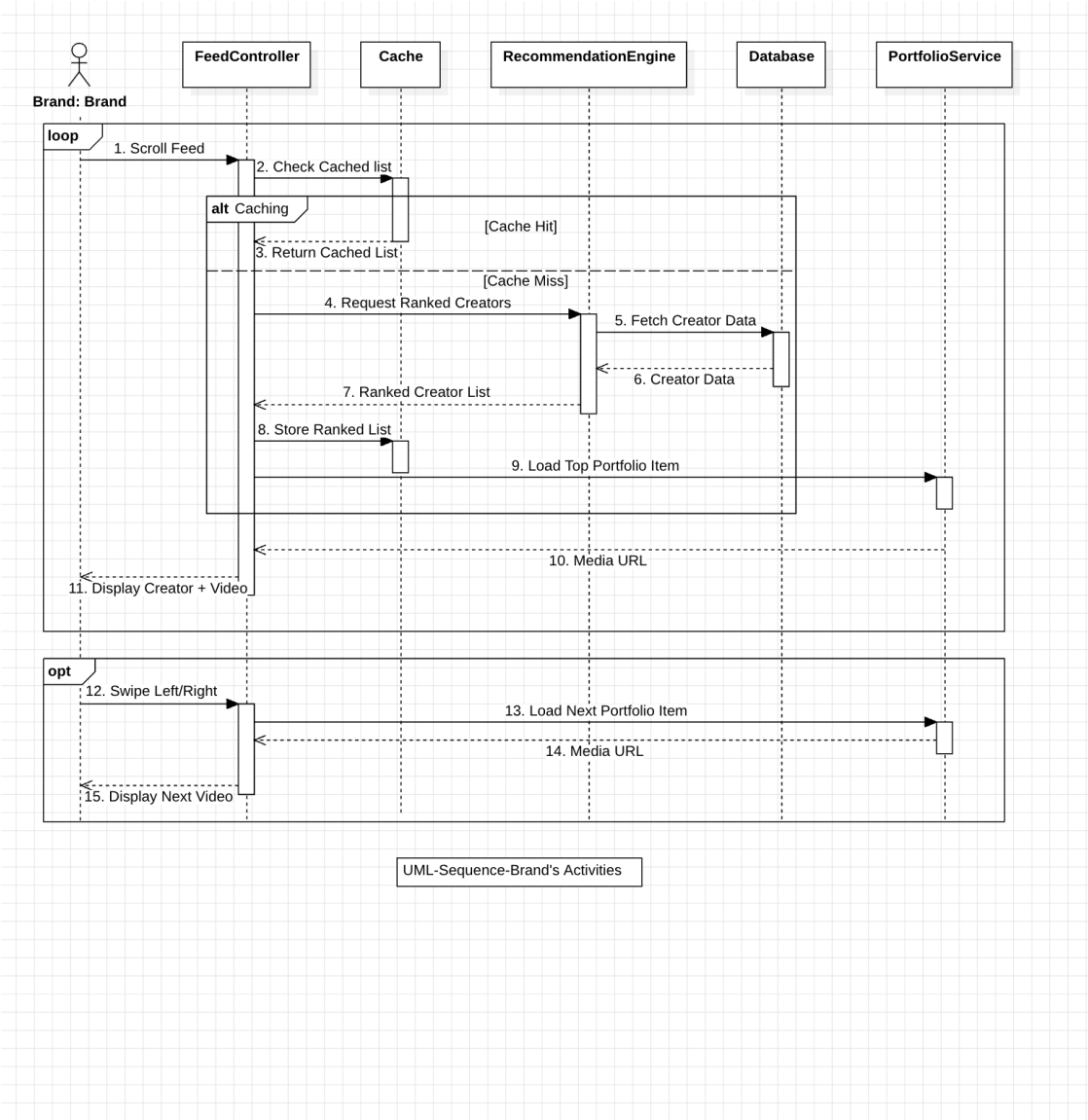
UML DIAGRAM: “ACTIVITY- PROFILE SETUP”



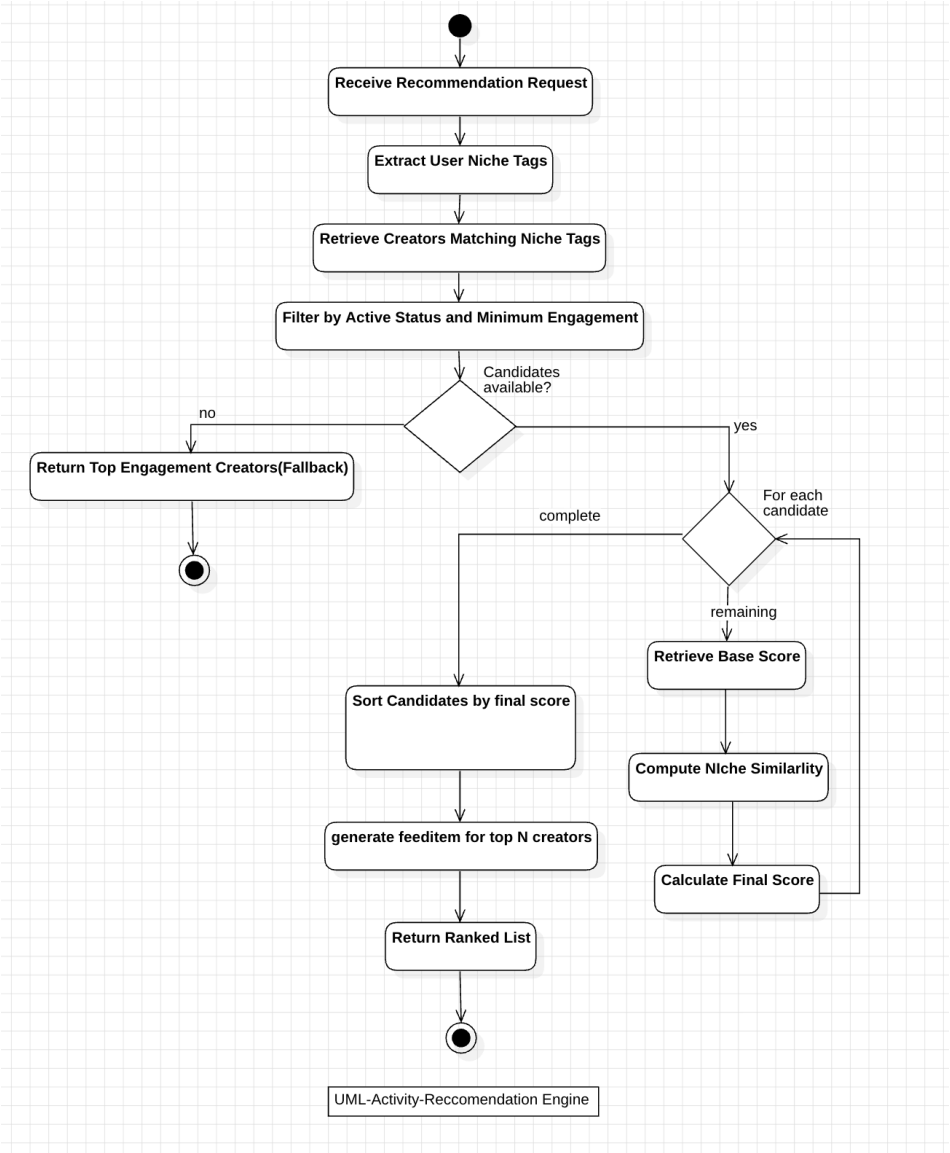
UML DIAGRAM: “ACTIVITY- CREATOR’S ACTIVITIES”



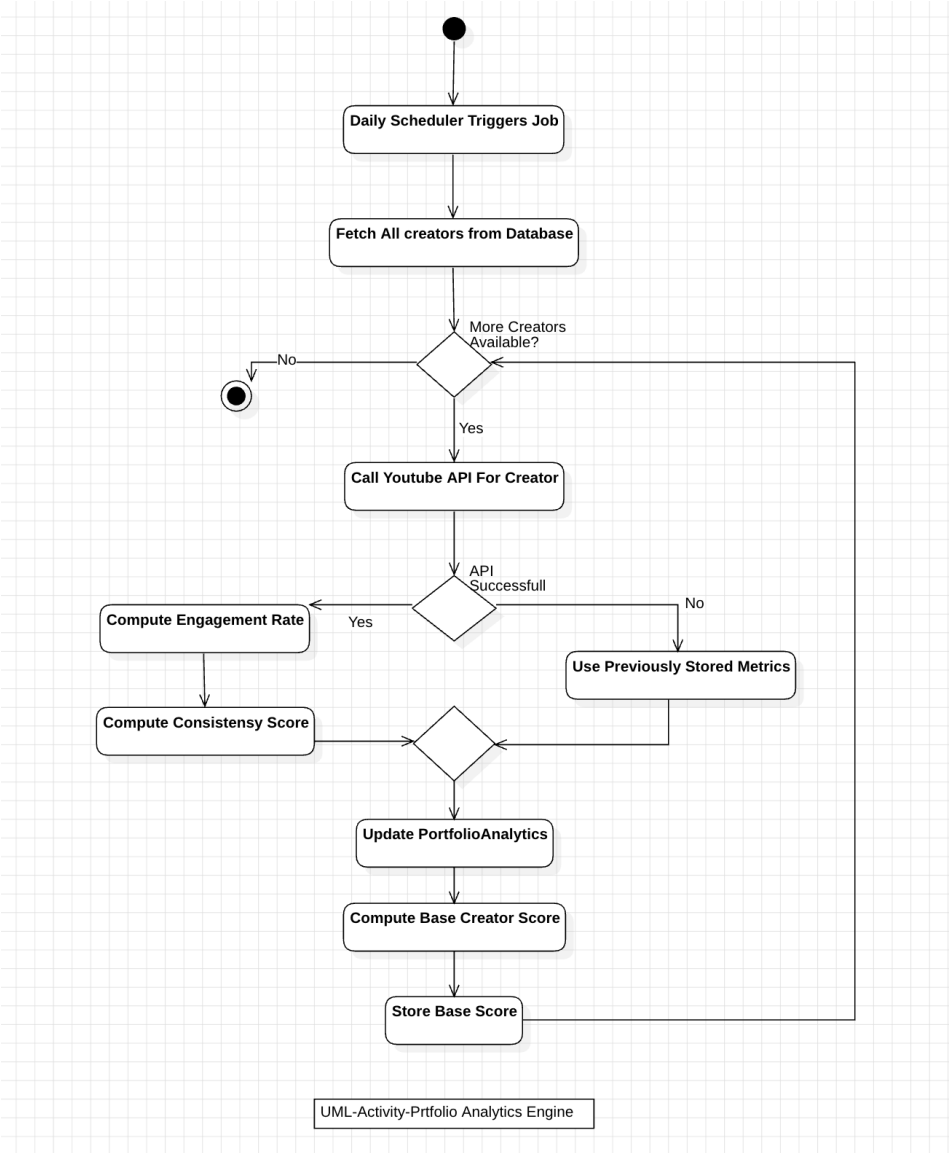
UML DIAGRAM: “SEQUENCE- BRAND’S ACTIVITIES”



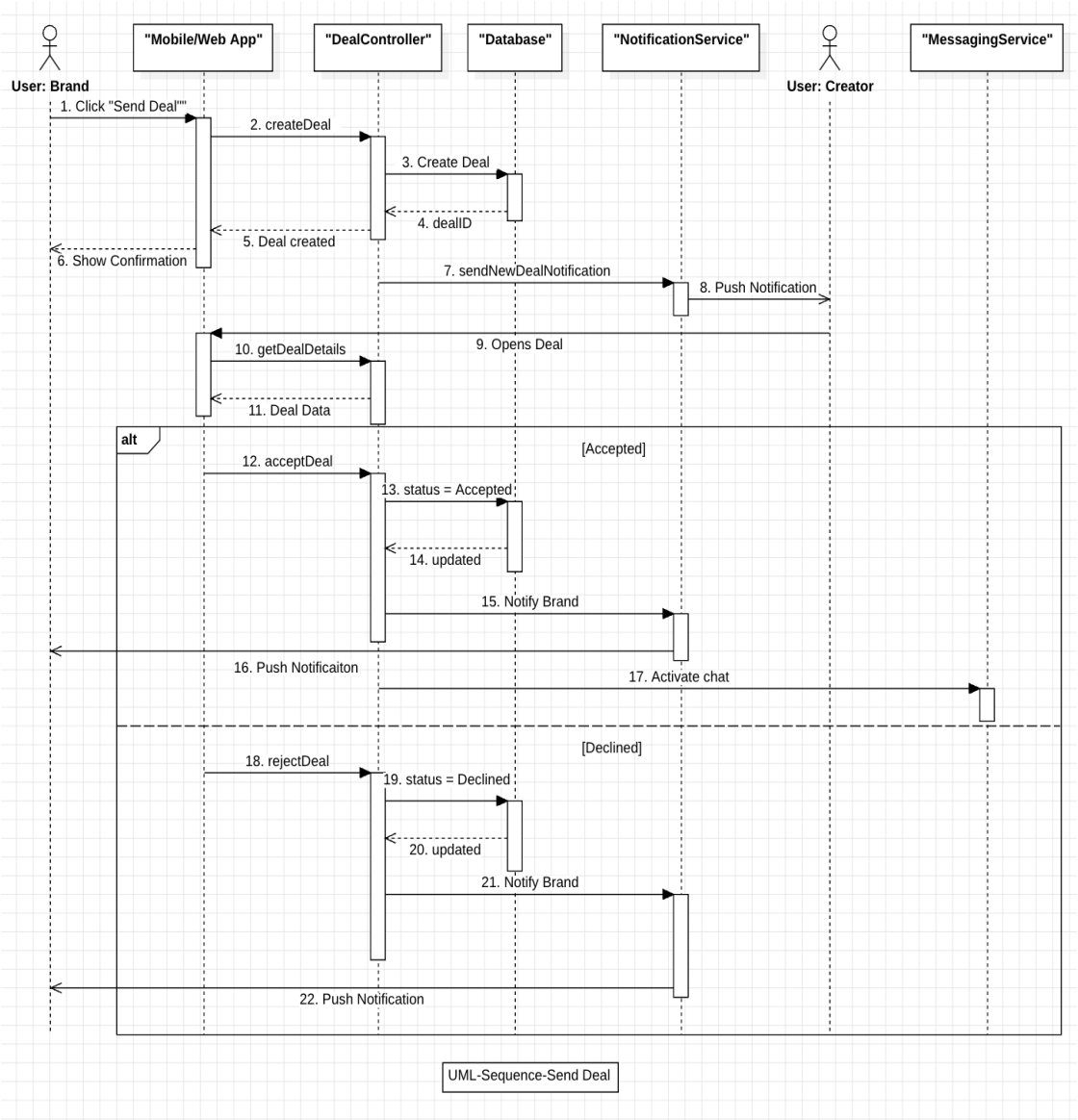
UML DIAGRAM: “ACTIVITY- RECOMMENDATION ENGINE”



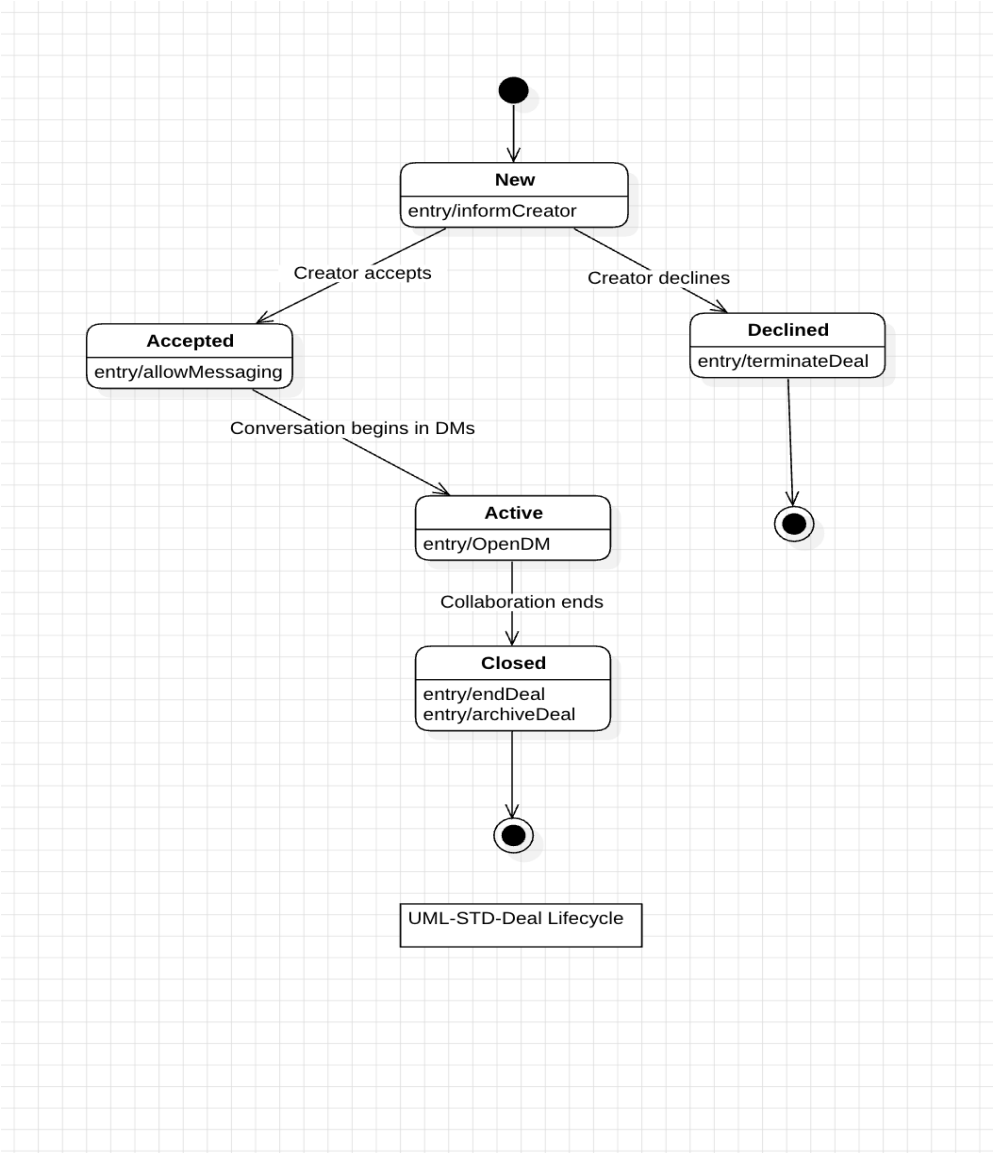
UML DIAGRAM: “ACTIVITY- PORTFOLIO ANALYTICS ENGINE”



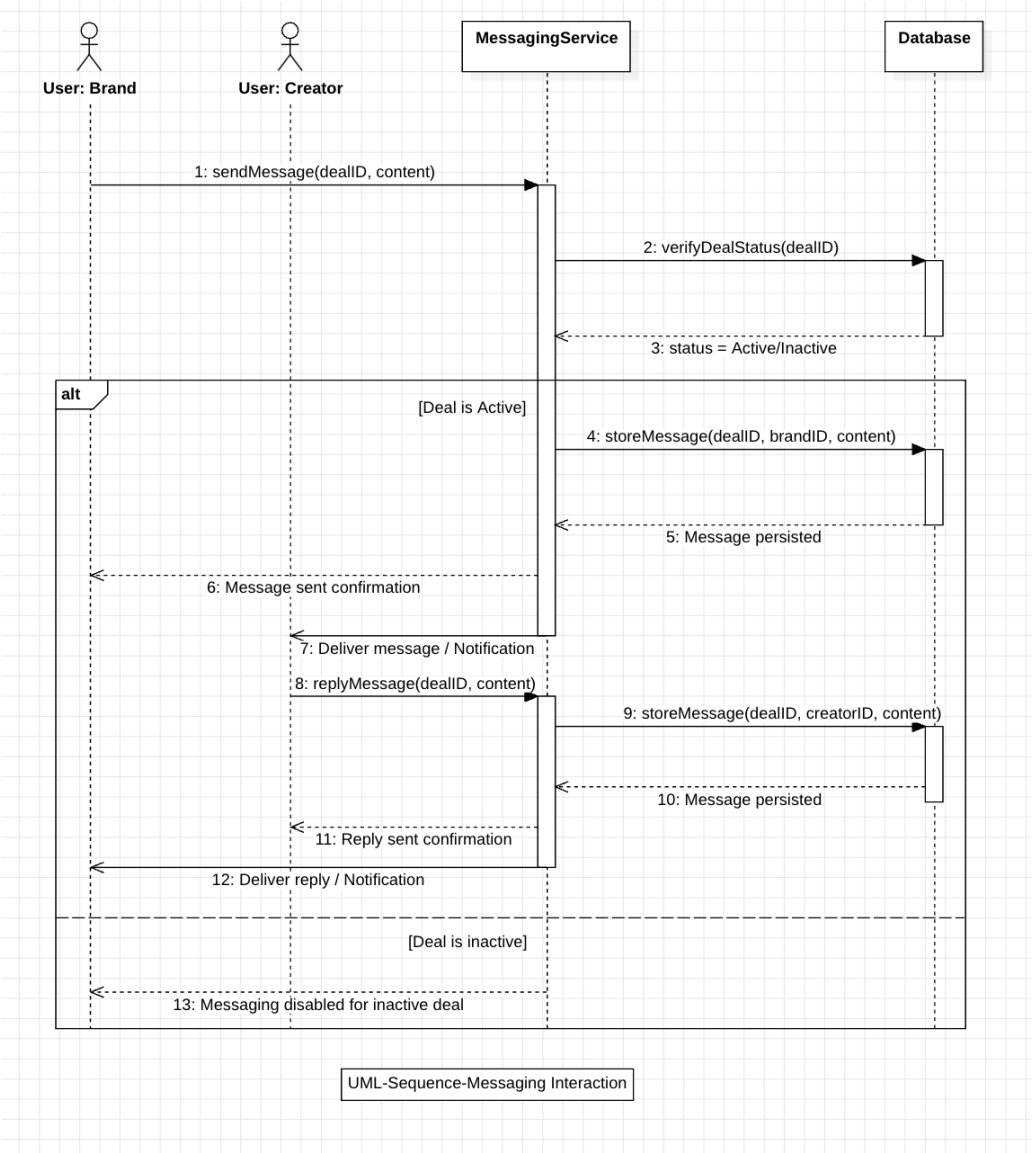
UML DIAGRAM: “SEQUENCE- SEND DEAL”



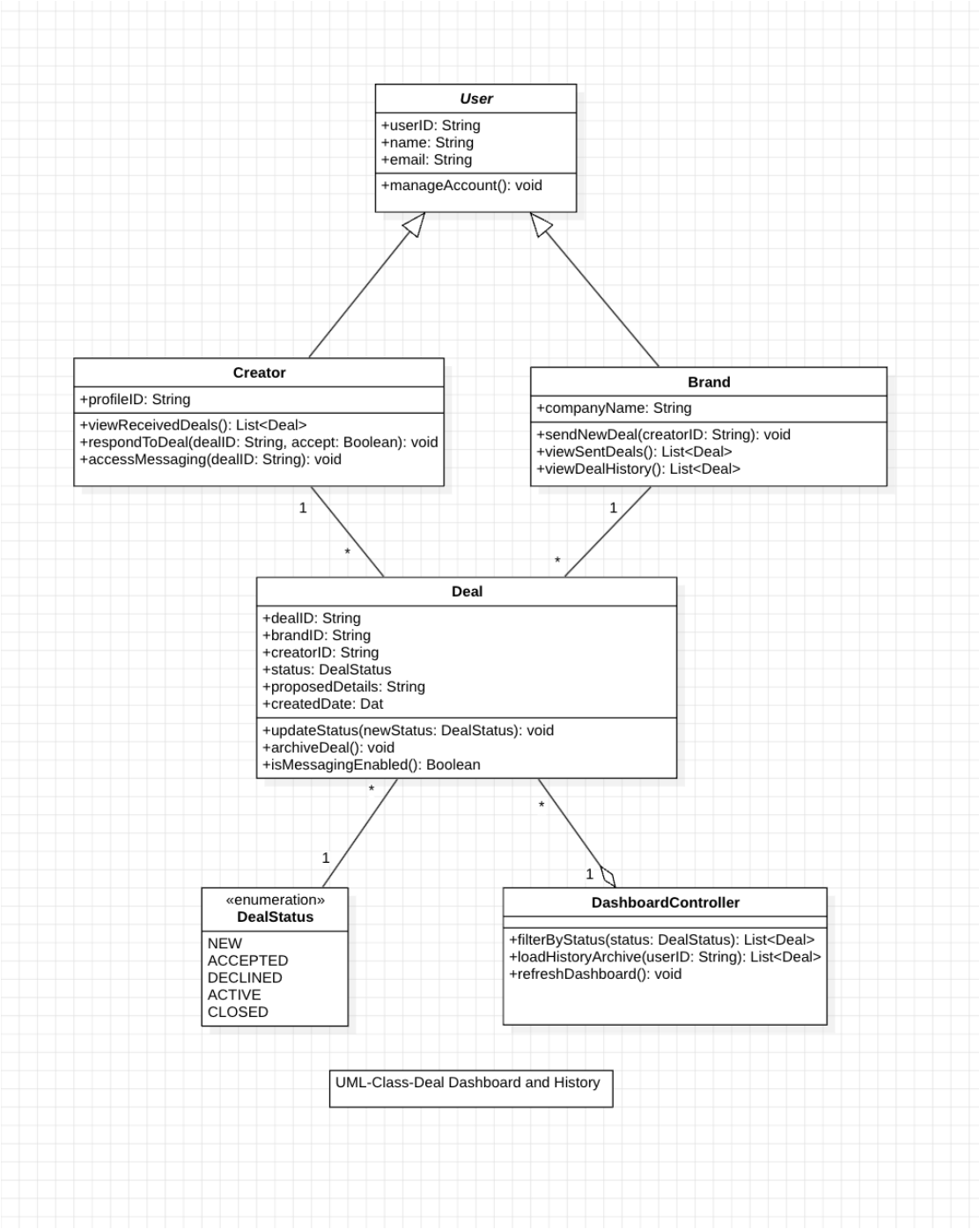
UML DIAGRAM: “STD- DEAL LIFECYCLE”



UML DIAGRAM: “SEQUENCE- MESSAGING INTERACTION”



UML DIAGRAM: “CLASS-DEAL DASHBOARD AND HISTORY”



5.3 Conceptual Data Model

NONE AVAILABLE AT THIS TIME

5.4 Screen Shots

NONE AVAILABLE AT THIS TIME

5.5 Test Plan

Test plan will be provided at a later stage of the project.

6. Assumptions and Constraints

6.1 ASSUMPTIONS

The following is a list of assumptions regarding the operational environment and user behavior that guide the development of the Creator Brand Collaboration System:

- Network and Device Connectivity: It is assumed that all users will operate the application on Android devices with a stable internet connection. Because the system relies on the YouTube Data API and real-time messaging, continuous connectivity is required. It is further assumed that users have push notifications enabled to receive time-sensitive deal updates.

- Data Integrity and Verification: The system assumes users will provide truthful profile information and valid portfolio media. A key assumption is that Brands will possess and upload a valid Business License or corporate profile for verification purposes. The system assumes these documents are authentic and legally represent the registering entity.

- API Accessibility and Permissions: It is assumed that the YouTube Data API will remain accessible and that its current quota limits are sufficient for prototype testing. Furthermore, it is assumed that Creators will grant the necessary OAuth 2.0 permissions to allow the system to fetch their channel metrics (subscribers, views, and content metadata).

- Age and Legal Capacity: All registered Creators and Brand representatives are assumed to be at least 18 years of age and legally authorized to enter into professional collaboration agreements. The system does not perform age-gate verification beyond the user’s self-declaration.

- Off-Platform Financial Settlements: The system assumes that Brands and Creators will independently manage all financial compensation and tax obligations outside of the application. The system provides the workflow for the "Deal," but assumes the actual transfer of funds occurs via third-party banking or payment services.

- Modular Extensibility: It is assumed that the internal data architecture is designed generically enough to eventually accommodate additional social media platforms (Instagram, TikTok, etc.) once their respective APIs are integrated in future versions.

6.2 CONSTRAINTS

The following constraints outline the technical, temporal, and resource-based limitations that restrict the design and implementation of the system:

- **Platform Specificity:** The software is strictly constrained to the Android mobile operating system. No iOS or web-browser equivalents will be developed during this lifecycle, meaning all UI/UX decisions are tailored specifically to Android Material Design standards.

- **Localized Persistence Layer:** The system will utilize SQLite as the primary database. This confines data persistence to the local device storage rather than an enterprise-level cloud server, which limits the ability to sync data across multiple devices for a single user account.

- **Academic Timeline and Team Experience:** The project operates under a non-negotiable deadline of April 28, 2026. This timeframe, combined with the team's ongoing learning curve in native Android development, restricts the implementation to core functionalities and prevents the inclusion of highly polished, market-ready "fluff" features.

- **API Implementation Scope:** While the system is designed to be multi-platform, the active integration is strictly constrained to the **YouTube Data API** for this project. No data will be pulled from other social media sources in this version.

- **Experimental AI Matchmaking:** The Recommendation Engine is constrained by the processing power of the local Android device. While the team will integrate AI-enhanced logic to assist in matchmaking, it will be a lightweight, mobile-optimized implementation rather than a heavy, cloud-based neural network.

- **Data Privacy of Legal Documents:** The requirement for Brands to upload Business Licenses introduces a security constraint. The system must ensure these files are stored in a protected directory within the SQLite environment to prevent unauthorized access.

6.3 OUT OF SCOPE MATERIAL

The following features are explicitly declared out of scope to ensure project delivery by the April 28th deadline:

- **In-App Financial Transactions:** Any form of payment processing, escrow services, or integration with payment gateways (e.g., Stripe, PayPal) to facilitate the transfer of funds between Brands and Creators.

- **Legal Contract Management:** The automated generation, digital signing (e.g., DocuSign), or legal enforcement of binding contracts, NDAs, or Terms of Service agreements beyond the digital "Accept/Decline" deal workflow.

- **Full Social Media Integration Suite:** The active integration of APIs for **Instagram, TikTok, Twitch, or X (Twitter)** is excluded from this phase; the system remains a YouTube-centric prototype.

- **Automated Government Verification:** Real-time validation of uploaded Business Licenses against government registries or Secretary of State databases is out of scope; verification is handled as a system status change based on document upload.

- **Advanced Cloud Infrastructure:** The development of a web portal for desktop users, a centralized administrative moderation panel, or long-term cloud-based data warehousing for multi-year analytics tracking.

- **Post-Deployment Support:** Any maintenance, software updates, or server-side support provided after the final project submission and product demo.

7. Delivery and Schedule

{List all tasks/milestones from start of the project to the end with specific dates for both Anticipated Start & End Dates

Task/Milestone Description	Anticipated Start Date	Anticipated End Date	Status {Complete, In Progress, To Be Completed-TBC}	Comments
Prepare UML diagrams	2/6/2026	2/19/2026	Complete	Deliverable è UML document
SRA document (Includes project objectives, Requirements and UML diagrams)	2/24/2026	3/26/2026	In Progress	Deliverable will be the SRA document. All stakeholders agree on the content of the SRA by signing in section 8.
Application Homepage design	2/20/2026	3/5/2026	TBC	UI - Team
Login and Registration	3/6/2026	3/15/2026	TBC	User Authentication - Team
Profile Setup	3/16/2026	3/22/2026	TBC	Creator and Brand Profiles - Team
Creator's Activities	3/23/2026	3/30/2026	TBC	Browse Creators - Team
Brand's Activities	3/23/2026	3/30/2026	TBC	Find Creators for Collabs - Team
Recommendation Engine	3/27/2026	4/5/2026	TBC	Matching algorithm - Team
Portfolio Analytics Engine	3/29/2026	4/8/2026	TBC	Score calculation - Team
Send Deal	3/31/2026	4/7/2026	TBC	Proposal workflow - Team
Deal Lifecycle	4/3/2026	4/12/2026	TBC	Status transitions - Team
Messaging Interaction	4/5/2026	4/16/2026	TBC	Secure chat - Team

Deal Dashboard and History	4/10/2026	4/19/2026	TBC	Tracking deals - Team
Test Plan Delivery	3/31/2026	4/16/2026	TBC	Deliverable will be the Test plan document.
User Manual	4/17/2026	4/28/2026	TBC	Documentation
Final Milestone: project delivery	4/28/2026	4/28/2026	TBC	Deliverable will be the final project binder plus product demo

8. Stakeholder Approval Form

Stakeholder Name	Stakeholder Role	Stakeholder Comments	Stakeholder Approval Signature and Date
Bahram Khalili	Client		
Sun Lin (Robin)	Client Project Manager		
Mehruba Tanzid Ontika	Developer		M.T.Ontika 03-25-2026
Nievanik Thapa Shrestha	Developer		N T Shrestha 03-25-2026
Thongum Athoiba	Developer		Thongum Athoiba 03-25-2026
Heathika Karicharla	Developer		Kheathika, 03-25-2026

Appendix:

None.